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Generate a lecture scribe intended to serve as reference material for exam preparation.

Use only the provided material. The scribe must strictly follow the content, structure, and scope of the provided materials and must not introduce any information, examples, explanations, or interpretations that are not present in the context.

The lecture scribe should faithfully reconstruct what was taught in the lecture so that a student can rely on it while preparing for a reading-based or closed-notes exam.

Include the following elements only when they appear in the provided context:
• Definitions and notation
• Assumptions and conditions
• Statements of theorems or formal results
• Proofs or proof sketches
• Mathematical derivations
• Worked examples with intermediate steps

Organize the scribe clearly using appropriate sections and headings so that the logical flow of the lecture is preserved and concepts can be easily reviewed for revision.

Maintain a formal and precise academic tone suitable for exam reference material.

Do not invent new material, do not simplify or reinterpret concepts, and do not include explanations or examples that are not explicitly present in the provided context.

Below is the **lecture scribe reconstructed strictly from the provided slides**. The content follows the same structure and includes only the material present in the lecture slides of **CSE400 – Lecture 4: Joint Probability and Conditional Probability**.
Source: L4

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability

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Date: January 15, 2026

L4

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1. Engineering Applications of Probability

Examples of engineering systems where probability is applied include:

1. **Speech Recognition System**
   A vocabulary such as *Hello, Yes, No, Bye* can be represented using waveform templates. Different speakers (male, female, child) produce different waveform patterns for the same word.
2. **Radar System**
   Radar signals are transmitted and reflected from objects such as aircraft. The received signal is processed to determine object detection.

3. **Communication Network**
   Data is transmitted from a **source** to a **destination** through multiple network nodes and devices.

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2. Introduction to Probability Theory

2.1 Experiments, Sample Space, and Events

Definition: Experiment (E)

An **experiment** is a procedure that produces some result.

Example:
Tossing a coin five times E_5.

Definition: Outcome (ξ)

An **outcome** is a possible result of an experiment.

Example:
One outcome of E_5:

$$\xi_1 = HHTHT$$

Definition: Event

An **event** is a set of outcomes of an experiment.

Example:
Let event C in experiment E_5 be:

$$C = \{\text{all outcomes consisting of an even number of heads}\}$$

Definition: Sample Space (S)

The **sample space** is the collection or set of all possible distinct outcomes of an experiment.

These outcomes are:

- **Mutually Exclusive**
  Example: In a coin flip, you can obtain heads or tails, but not both.
- **Collectively Exhaustive**
  All possible outcomes are included in the sample space.

Example: Flip a coin.

Properties of Sample Space

- S is the **universal set of outcomes** of an experiment.
- The sample space can be:
  - **Discrete**
  - **Countably infinite**
  - **Continuous**

Examples of Experiments

1. Flipping a fair coin once
2. Rolling a cubical die with numbered faces
3. Rolling two dice
4. Flipping a coin until a tails occurs

5. Random number generator with interval $[0, 1]$

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3. Axioms of Probability

Definition: Probability

Probability is a **measure of the likelihood of events** or a function of an event that produces a numerical quantity measuring the likelihood of that event.

Axioms

The following statements are taken as **self-evident and require no proof**.

1. For any event A:

$$0 \leq Pr(A) \leq 1$$

2. If S is the sample space:

$$Pr(S) = 1$$

3. If A and B are mutually exclusive ($A \cap B = \emptyset$):

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

4. For mutually exclusive sets A_i:

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i)$$

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4. Corollary from Probability Axioms

Let A be the **null event** for all values of i greater than n.

Corollary 2.1

For M mutually exclusive events:

$$Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} Pr(A_i)$$

- Axiom 3 is equivalent to Corollary 2.1 when the sample space is finite.
- The generality of Axiom 3 is necessary when the sample space contains infinitely many points.

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5. Propositions from Probability Axioms

Proposition 2.1

$$Pr(A^c) = 1 - Pr(A)$$

Proposition 2.2

If $A \subset B$:

$$Pr(A) \leq Pr(B)$$

Proposition 2.3

For any sets A and B:

$$Pr(A \cup B) = Pr(A) + Pr(B) - Pr(A \cap B)$$

Proposition 2.4 (Inclusion–Exclusion Principle)

$$Pr(A_1 \cup A_2 \cup ... \cup A_M) = \sum_{i=1}^{M} Pr(A_i) - \sum_{i_1<i_2} Pr(A_{i_1} A_{i_2}) + ... + (-1)^{M+1} Pr(A_1 A_2 ... A_M)$$

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6. Assigning Probabilities

Classical Approach

Example 2.6

Coin flipping experiment:

Compute

$$Pr(H), \quad Pr(T)$$

Example 2.7

Dice rolling experiment:

Compute

$$Pr(\text{even number is rolled})$$

Example 2.8

Two dice are rolled.

Let event A:

Sum of the two dice equals five.

Compute:

$$Pr(A)$$

Relative Frequency Approach

Example: Dice experiment where event A = sum of two dice equals five.

Simulation results:

n	1000	2000	3000	4000	5000	6000	70
n_A	96	200	314	408	521	630	75
n_A/n	0.096	0.100	0.105	0.102	0.104	0.105	0.1

Inference

- To obtain the exact probability of an event, the experiment must be repeated **infinitely many times**.
- Many random phenomena of interest are **not repeatable**.

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7. Joint Probability

Motivation

Events are not always mutually exclusive.

We are interested in the probability of the **intersection of events A and B**.

Notation

Joint probability is denoted as:

$$Pr(A, B) \quad \text{or} \quad Pr(A \cap B)$$

Multiple Events

For events $A_1, A_2, ..., A_M$:

$$Pr(A_1, A_2, ..., A_M)$$

Calculation Approaches

Classical Approach

1. Express events A and B in terms of atomic outcomes.
2. Identify outcomes common to both.
3. Compute the probability of these outcomes.

Relative Frequency

Let $n_{A,B}$ be the number of trials where both A and B occur.

$$Pr(A, B) = \lim_{n \to \infty} \frac{n_{A,B}}{n}$$

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8. Example 1: Card Deck

Consider a deck of **52 playing cards**.

Define events:

- A: Red card selected
- B: Number card selected (Ace counted as number)
- C: Heart card selected

Find:

$$Pr(A), Pr(B), Pr(C), Pr(A, B), Pr(A, C), Pr(B, C)$$

Solution

Individual Probabilities

$$Pr(A) = \frac{26}{52} = \frac{1}{2}$$

$$Pr(B) = \frac{40}{52} = \frac{5}{26}$$

$$Pr(C) = \frac{13}{52} = \frac{1}{4}$$

Joint Probabilities

$$Pr(A, B) = \frac{20}{52} = \frac{5}{13}$$

$$Pr(A, C) = \frac{13}{52} = \frac{1}{4}$$

$$Pr(B, C) = \frac{10}{52}$$

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9. Example 2: Costume Party

Alex has the following clothing:

Tops:

- 3 white T-shirts
- 1 radioactive green cape

Bottoms:

- 2 flannel dinosaur pajama pants
- 4 polka-dot boxer shorts

Step 1: Analyze Tops

Total tops:

$$3 + 1 = 4$$

$$Pr(\text{Cape}) = \frac{1}{4}$$

Step 2: Analyze Bottoms

Total bottoms:

$$2 + 4 = 6$$

$$Pr(\text{Boxers}) = \frac{4}{6} = \frac{2}{3}$$

Step 3: Joint Probability

$$Pr(\text{Cape} \cap \text{Boxers}) = Pr(\text{Cape}) \times Pr(\text{Boxers})$$

$$= \frac{1}{4} \times \frac{4}{6} = \frac{1}{6}$$

Correct Option: **C**

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10. Conditional Probability

Motivation

The occurrence of one event may depend on another event.

Definition

The probability of event A given event B:

$$Pr(A|B) = \frac{Pr(A, B)}{Pr(B)}$$

where

$$Pr(B) > 0$$

Product Rule

$$Pr(A,B) = Pr(A|B)Pr(B) = Pr(B|A)Pr(A)$$

Conditional Probability of Three Events

$$Pr(A,B,C) = Pr(C|A,B)Pr(B|A)Pr(A)$$

Chain Rule for M Events

$$Pr(A_1, A_2, ..., A_M) = Pr(A_M|A_1, ..., A_{M-1}) \times ... \times Pr(A_2|A_1) \times Pr(A_1)$$

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11. Example 3: Cards Without Replacement

Two cards are selected without replacement.

Define:

- A: First card is a spade
- B: Second card is a spade

Find:

$$Pr(B|A)$$

Initial State

- Total cards = 52
- Spades = 13

After Event A

One spade removed.

Remaining:

- Total cards = $52 - 1 = 51$
- Spades = $13 - 1 = 12$

Calculation

$$Pr(B|A) = \frac{12}{51}$$

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12. Example 4: Game of Poker

Five cards are dealt.

A **flush** means all cards are from the same suit.

Part 1: Flush in Spades

$$P(\text{1st spade}) = \frac{13}{52}$$

$$P(\text{2nd spade}) = \frac{12}{51}$$

$$P(\text{3rd spade}) = \frac{11}{50}$$

$$P(\text{4th spade}) = \frac{10}{49}$$

$$P(\text{5th spade}) = \frac{9}{48}$$

$$P(\text{Spade Flush}) = \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} \cdot \frac{10}{49} \cdot \frac{9}{48}$$

Part 2: Any Flush

There are four suits:

Spades, Hearts, Diamonds, Clubs.

Since events are mutually exclusive:

$$P(\text{Any Flush}) = 4 \times P(\text{Spade Flush})$$

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13. Example 5: The Missing Key

Bob is **80% certain** the key is in his jacket.

- Left pocket: $P(L) = 0.4$
- Right pocket: $P(R) = 0.4$
- In jacket: $P(K) = 0.8$

Goal:

$$P(R|L^c)$$

Probability that the key is in the **right pocket given it is not in the left pocket**.

Calculation

$$P(R|L^c) = \frac{P(R \cap L^c)}{P(L^c)}$$

Since R implies L^c:

$$P(R \cap L^c) = P(R)$$

$$P(L^c) = 1 - P(L)$$

Therefore

$$P(R|L^c) = \frac{0.4}{1 - 0.4} = \frac{0.4}{0.6} = \frac{2}{3}$$

Correct Option: **C**

If you want, Datt, I can also **convert this into a perfectly formatted PDF/notes version for studying (much shorter and exam-optimized)** so you can revise **this entire lecture in 10–15 minutes before exams.**

Sources

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